

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 04

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Find the range of values of x for which $x^2 - 7x + 10 < 0$ and $x^2 - 4 > 0$.

A $2 < x < 5$

B $-2 < x < 2$

C $x > 5$

D $x < -2$

E $2 < x < 5$ or $x < -2$

F no values

2 Find the equation of the normal to the curve $y = \sqrt{x}$ at the point $(4, 2)$.

A $y = -4x + 16$

B $y = -4x + 18$

C $y = -4x + 2$

D $y = \frac{1}{4}x + 1$

E $y = -\frac{1}{4}x + 3$

F $y = 4x - 14$

3 Find the highest common factor of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

A 18

B 6

C 12

D 36

E 72

F 108

- 4 The gradient of a curve at (x, y) is $3\sqrt{x}$ and the curve passes through $(4, 20)$. Find the y-intercept of the curve.
- A** 12
- B** 20
- C** 4
- D** 8
- E** -4
- F** 0

- 5 Evaluate $\sum_{k=1}^{10} 3 \times 2^k$.
- A** 6141
- B** 3069
- C** 3072
- D** 3066
- E** 6144
- F** 6138

- 6** The distance-time graph of a journey is a straight line from $(0, 0)$ to $(2, 60)$, then horizontal to $(3, 60)$, then a straight line to $(5, 0)$ (time in hours, distance from home in km). What was the average speed for the whole journey?
- A** 0 km/h
 - B** 12 km/h
 - C** 24 km/h
 - D** 20 km/h
 - E** 30 km/h
 - F** 40 km/h

- 7 A ladder of length 6.5 m leans against a vertical wall with its foot 2.5 m from the wall on horizontal ground. How far up the wall does it reach?
- A 7 m
 - B 5 m
 - C 4 m
 - D 5.5 m
 - E 6 m
 - F $\sqrt{48.5}$ m

8 The line $ax + by = 12$ has x -intercept 3 and y -intercept -4 . Find $a - b$.

A 12

B 7

C -1

D 0

E -7

F 1

9 In the expansion of $(1 + x)^n$, the coefficients of x^4 and x^5 are equal. Find n .

A 20

B 5

C 11

D 9

E 10

F 8

10 Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

- A** 270
- B** 1080
- C** 2160
- D** 720
- E** 540
- F** 360

- 11** The line $y = mx + c$ passes through $(2, 7)$. If m is increased by 1 and c is decreased by 2, the new line passes through $(2, k)$. Find k .

- A** 6
- B** 7
- C** 5
- D** 9
- E** 8
- F** 10

12 Evaluate $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$.

- A** $\frac{2}{3}$
- B** $\frac{3}{7}$
- C** $\frac{12}{49}$
- D** $\frac{7}{12}$
- E** $\frac{2}{24}$
- F** 1

- 13** The line $y = mx$ meets the curve $y = x^3 - x$ at three distinct points. Find the complete set of values of m .
- A** $m < -1$
- B** $-1 < m < 0$
- C** $m > 1$
- D** $m > 0$
- E** $m > -1$
- F** all real m

- 14** Find the coordinates of the stationary points of $y = x^3 - 3x^2 - 9x + 5$ and state their nature.
- A** maximum at $(1, -6)$, minimum at $(-3, -22)$
 - B** minimum at $(-1, 10)$, maximum at $(3, -22)$
 - C** maximum at $(-1, 10)$, minimum at $(3, -22)$
 - D** maximum at $(-1, 0)$, minimum at $(3, -32)$
 - E** maximum at $(-1, 10)$ only

- 15** A vertical pole BT stands on horizontal ground. From a point A on the ground the angle of elevation of T is 30° ; from a point C on the ground, 20 m closer to the pole than A and in line with A and B , the angle of elevation is 60° . Find the height of the pole.

- A** 10 m
- B** $20\sqrt{3}$ m
- C** 15 m
- D** $10\sqrt{3}$ m
- E** $\frac{20}{\sqrt{3}}$ m
- F** 30 m

- 16** For which value of k does $x^2 - x + k$ divide exactly into $x^3 - 2x^2 + 3x - 4$? (Assume such a value exists.)
- A** -2
 - B** 3
 - C** 4
 - D** 2
 - E** 1
 - F** no such value exists

17 The first three terms of a geometric sequence are x , $x + 6$ and $x + 15$. Find the fourth term.

- A** $\frac{81}{2}$
- B** $\frac{125}{2}$
- C** $\frac{75}{2}$
- D** $\frac{225}{4}$
- E** $\frac{125}{4}$
- F** 45

- 18** The equation $x^2 + ax + b = 0$ has roots 3 and -5 . The equation $x^2 + bx + a = 0$ has roots p and q . Find $p^2 + q^2$.
- A** 229
- B** 195
- C** 4
- D** 221
- E** 225
- F** 255

19 Which of the following is equal to $\frac{x^2\sqrt{x}}{\sqrt[3]{x}}$ for $x > 0$?

A x^3

B $x^{\frac{5}{2}}$

C $x^{\frac{7}{3}}$

D $x^{\frac{11}{6}}$

E $x^{\frac{17}{6}}$

F $x^{\frac{13}{6}}$

- 20** The perimeter of a sector of radius r and angle θ radians is 20, and its area is 16. Find the possible values of r .
- A** 4 only
 - B** 8 only
 - C** 1 or 16
 - D** 2 or 8
 - E** 4 or 8
 - F** 2 or 4

END OF TEST